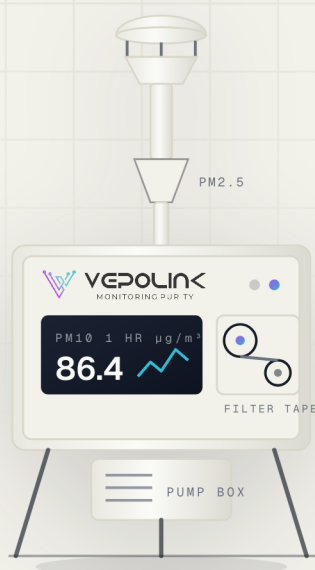


CONTINUOUS AMBIENT AIR QUALITY MONITORING

PM10 & PM2.5 *Beta* Attenuation Monitor

Reference-grade hourly particulate mass by beta ray attenuation — measured in-line, in-situ, in a lightweight shelter that needs no air conditioning and no walk-in cabin.



PM10 · 2.5

GRAVIMETRIC-EQUIVALENT MASS

16.7 L/min

REGULATED SAMPLE FLOW

1 hour

STANDARD TIME RESOLUTION

1hrSTANDARD TIME
RESOLUTION**16.7**L/minREGULATED SAMPLE
FLOW RATE**45**daysFILTER TAPE LIFE
AT 1 SAMPLE /
HOUR**18**kgHEAVIEST SINGLE
MODULE

01 OVERVIEW

Continuous particulate mass, *measured not modelled.*

A continuous, real-time beta attenuation monitor for ambient PM10 and PM2.5.

Particulate is collected on a glass fibre filter tape and its mass determined directly from the attenuation of beta radiation — the nationally and internationally recognised technique for regulatory particulate measurement.

An in-line sampling geometry measures beta attenuation across the filter media while sampling continues, so readings accumulate through the whole hour rather than in short windows. That improves both sensitivity and temporal resolution, and lets the station run unattended, 24 hours a day, every day.

REPORTED CHANNELS

- PM10
- PM2.5
- HOURLY MASS
- 24-HR AVERAGE
- SAMPLE FLOW
- AMBIENT TEMP
- RELATIVE HUMIDITY
- BAROMETRIC PRESSURE
- FILTER TAPE STATUS
- DIAGNOSTICS



In-Situ Beta Measurement

Mass is measured where the sample is collected — high accuracy with minimal measurement artefacts and reduced background determination.



No Dryer, No Air Conditioning

Sample is held slightly above ambient temperature, preventing moisture artefacts and semi-volatile loss without costly conditioning hardware.



Lightweight Field Shelter

No module exceeds 18 kg, so one person can carry and site the system where a permanent structure or walk-in shelter is unavailable.



Remote & Cloud Operation

Advanced communications and diagnostics allow remote configuration, cloud reporting and early warning of drift or flow deviation.

02 MEASUREMENT PRINCIPLE

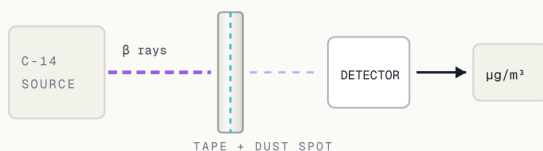
How beta attenuation *weighs air*.

Beta rays lose energy in proportion to the mass they pass through. Collect particulate on a filter tape, measure the beta signal before and after, and the difference is a direct measurement of collected mass — no assumption about particle shape, colour or refractive index.

BETA ATTENUATION

Mass by Radiation Loss

C-14 SOURCE · FILTER TAPE · DETECTOR



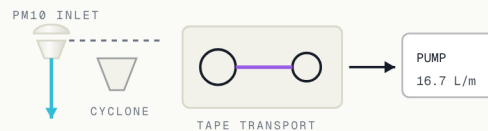
A low-activity carbon-14 source emits beta particles through the filter tape onto a detector. As dust accumulates, the count rate falls; the attenuation gives collected mass, which the flow volume converts to concentration.

µg/m³ from measured mass

SIZE SELECTION

Inlet Head & Cyclone

PM10 HEAD · SHARP-CUT CYCLONE FOR PM2.5



Air enters through a PM10 size-selective inlet at a regulated 16.7 L/min. For PM2.5, a sharp-cut cyclone removes coarse fraction before the tape. No flow splitters are used, which simplifies the flow path and maintenance.

Active volumetric flow control

SAMPLE CONDITIONING

In-Situ Thermal Control

NEARLY CONSTANT ABOVE AMBIENT



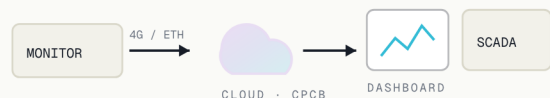
The beta measurement is held at a nearly constant temperature slightly above ambient. This prevents excess moisture and the loss of semi-volatile particulate, minimising error without expensive membrane dryers.

High dew point performance

DATA PATH

Reporting & Integration

ETHERNET · RS-485 · 4G · ANALOG



Hourly and averaged values are logged internally and pushed to the cloud or to a regulatory server, with MODBUS and analog outputs for local SCADA. Diagnostics travel with the data.

Remote operation supported

03 MEASUREMENT CYCLE

One hour, *four steps*.

The in-line geometry means sampling and counting happen together, so the instrument spends the hour measuring rather than switching between collection and reading.

STEP 01 Fresh tape advance A clean section of glass fibre tape is positioned in the sampling head and a reference beta count is taken through the blank media.	STEP 02 Continuous sampling Ambient air is drawn through the tape at a regulated 16.7 L/min while the beta measurement continues across the accumulating deposit.	STEP 03 Attenuation compare The loaded count rate is compared against the reference. Attenuation converts to collected mass, corrected for temperature and pressure.	STEP 04 Report & repeat Mass divided by the measured sample volume gives concentration in $\mu\text{g}/\text{m}^3$. The value is logged, transmitted, and the cycle restarts.
--	---	--	--

WHY IN-LINE GEOMETRY

Instruments that alternate between sampling and counting spend part of every cycle blind. Measuring beta attenuation across the media *while* particles are collected keeps the detector counting throughout the hour, improving sensitivity and giving a cleaner, more responsive hourly value.

SAMPLING INTERVAL

One hour is the standard reporting interval. Where ambient levels are low, longer integration improves the detection limit; where levels are high or an event is being tracked, shorter intervals can be configured to suit the site and the reporting requirement.

DATA QUALITY & TRACEABILITY

Every reported concentration carries the conditions it was measured under. Sample volume is corrected for ambient temperature and barometric pressure, flow is actively regulated and audited rather than assumed, and span foil verification against the mass calibration system ties the beta measurement back to a known reference. Where a value falls outside those bounds, it is flagged in the record instead of silently averaged away — so the data set handed to a regulator or a consultant can be defended line by line.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

Parameters	PM10 and PM2.5
Working Principle	Beta ray attenuation (BAM)
Sampling Geometry	In-line – measurement during collection
Measuring Range	0 – 10,000 µg/m ³
Resolution	0.1 µg/m ³
Time Resolution	1 hour standard · configurable
Sample Flow	16.7 L/min, actively regulated
Size Selection	PM10 inlet head · sharp-cut cyclone for PM2.5
Beta Source	Carbon-14, low activity, exempt quantity
Filter Media	Glass fibre tape · approx. 45 days at 1 sample/hour
Sample Conditioning	In-situ heating – no membrane dryer required

SHELTER DIMENSIONS

SYSTEM · COMMS · BUILD

Auxiliary Sensors	Ambient temperature, relative humidity, barometric pressure
Optional Inputs	Meteorological and additional sensor channels
Outputs	Ethernet · RS-232 / RS-485 MODBUS · analog · USB
Remote Access	Cloud reporting and remote operation
Data Logging	Internal logger with diagnostics
Calibration	Mass calibration system · span foil verification
Power Supply	230 VAC · 50 Hz
Air Conditioning	Not required
Shelter	Lightweight enclosure with removable pump box
Module Weight	No component above 18 kg
Setup Time	Under 10 minutes

INDICATIVE FRONT & SIDE ELEVATION · MM

05 INSTALLATION & UPKEEP

Sited in a morning, *run for a season.*

Per CPCB guidance for continuous ambient stations, the sampling inlet is placed in free-flowing air clear of obstructions — typically **3 to 10 m above ground level**, at least **1 m clear of any structure** in all directions, and away from local sources such as chimneys, vents and unpaved roads.

01 · SITING Rooftop, pad or tripod No walk-in shelter or civil foundation needed. A level pad, rooftop or tripod with clear inlet exposure is sufficient.	02 · HANDLING One-person install Every module is under 18 kg and the pump box lifts out, so the system can be carried, positioned and running in under ten minutes.	03 · POWER 230 VAC, no AC unit Because the shelter does not need air conditioning, site power demand and running cost stay low even in hot climates.
04 · FILTER TAPE Changed roughly monthly A tape roll lasts about 45 days at one sample per hour. Replacement is a simple reel change with no tools.	05 · VERIFICATION Flow, leak & span Periodic flow audit, leak check and span foil verification against the mass calibration system keep results traceable.	06 · DIAGNOSTICS Faults flagged remotely Flow deviation, tape end, detector health and temperature status are reported with the data so issues surface before they cost data.

OPERATING COST

Consumables are limited to filter tape and occasional inlet cleaning. With no membrane dryer, no flow splitters and no air conditioner, there is less to service, less to fail and less power drawn — the reasons a beta attenuation station is practical to run at many sites at once rather than only at a flagship one.

RELOCATION

Because the shelter is modular and no part needs two people to lift, moving a station is a planned afternoon rather than a project. Sites used for a construction phase, a baseline survey or a seasonal campaign can be decommissioned and re-established elsewhere without re-engineering the installation.

06 APPLICATIONS

Where it *works*.

Anywhere particulate has to be reported rather than estimated — regulatory networks, plant boundaries, project compliance and research campaigns all use the same measurement.

 Regulatory CAAQMS Networks	 Industrial Plant Boundaries	 Roadside & Traffic Corridors
 Construction & Demolition	 Mining & Quarrying	 Power & Thermal Stations
 Ports, Yards & Logistics	 EIA & Baseline Studies	 Remote & Campaign Sites

TYPICAL DEPLOYMENTS

Two patterns cover most projects: a permanent regulatory station reporting hourly PM10 and PM2.5 to a pollution control board server, or a relocatable unit used for boundary monitoring, baseline surveys and short campaigns. The portability of the shelter means the same instrument can serve both roles across a project's life.

ALONGSIDE GAS ANALYSERS

The particulate monitor is commonly installed as part of a fuller CAAQMS package — SO₂, NO_x, CO and ozone analysers, a calibration system and a meteorological mast sharing the same time base, data path and reporting server. Specified together, the station reports a complete air quality picture from one site record.

Frequently asked *questions.*

Q What is the working principle?

Beta ray attenuation. Particulate collected on a glass fibre filter tape attenuates beta radiation in proportion to its mass, giving a direct mass measurement rather than an optical inference.

Q Why beta attenuation for compliance work?

Beta attenuation monitoring is the recognised national and international technology for continuous regulatory measurement of particulate matter, and is accepted as equivalent to gravimetric reference methods.

Q How long does one measurement take?

One hour is standard. In low-pollution regions a fixed shorter collection at 16.7 L/min is often averaged instead; for Indian conditions the interval can be set to suit the site and expected loading.

Q How often is the filter tape changed?

Approximately every 45 days at one sample per hour. The change is a simple reel swap, and tape end is flagged in advance through diagnostics.

Q Does it need a Nafion or membrane dryer?

No. In-situ thermal control keeps the measurement slightly above ambient, which handles moisture and protects semi-volatiles without dryer hardware or its running cost.

Q Can it measure PM2.5 and PM10?

Yes. A PM10 size-selective inlet is standard; a sharp-cut cyclone ahead of the tape gives the PM2.5 fraction. Configuration is set to your reporting requirement.

Q Is the beta source a safety concern?

The carbon-14 source is very low activity and sealed within the instrument, in the exempt category for handling and transport. Routine operation requires no radiological licence for the user.

Q Does the shelter need air conditioning?

No. The integrated shelter operates without air conditioning, which reduces power consumption, simplifies maintenance and improves overall reliability.

Q How quickly can it be deployed?

No component weighs more than about 18 kg, so one person can carry the system. Typical setup from arrival to running data is under ten minutes.

Q Can data reach a pollution board server?

Yes. Ethernet, RS-485 MODBUS, analog outputs and cellular cloud reporting are supported, so data can go to a regulatory server, a plant SCADA, or both.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

Report PM with *confidence.*

Tell us the site, the fractions you need to report and who receives the data, and we'll configure the monitor — inlet, cyclone, communications and shelter to match. Pick how you'd like to start.

Talk to our engineers

01

Speak directly to the team that specifies, installs and calibrates beta attenuation stations. No call centre.

[+91 99538 86178](tel:+919953886178)

Request a system configuration

02

RECOMMENDED

Send your site and reporting requirement — we'll spec inlet, cyclone, communications, shelter and consumables.

sales@vepolink.com

Book a demonstration

03

See a running station and the reporting dashboard with our engineering team.

[Request a demo](#)